

Eduardo Hafemann

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Education

PhD in Mathematics 10/2023 – 09/2026

Universität Hamburg (UHH)

Supervisor: Prof. Dr. Melanie Graf.

Co-supervisor: Prof. Dr. Eleni-Alexandra Kontou (King's College London).

Thesis Project: Non-smooth Techniques in Mathematical Relativity.

Funding: Cluster of Excellence Quantum Universe at Universität Hamburg and DESY.

Msc Mathematics 08/2021 – 08/2023

Federal University of Santa Catarina - Florianópolis (UFSC)

Supervisor: Prof. Dr. Ivan Pontual Costa e Silva.

Dissertation Title: Geometry and Topology of Black Hole Horizons ([RI UFSC](#)).

Bsc Chemical Engineering 08/2016 – 08/2021

Federal University of Santa Catarina - Florianópolis (UFSC)

Award: Highest GPA in Chemical Engineering, Graduating Class of August 2021.

Work Experience

Chemical Engineering Internship & Consultant 08/2020 – 12/2022

Tubanharon Process Engineering

Supervisor: Dr. Luismar Marques Porto.

Activities: Mathematical modeling for soybean oil extractor and web development.

Department of Mathematics Research Internship 08/2019 – 08/2021

Supervisor: Prof. Dr. Fábio Junior Margotti.

Theme: Inverse problems and Electrical Impedance Tomography (EIT).

Published Work: Articles [\[4, 5, 6\]](#) and book [\[12\]](#).

Department of Physics Research Internship (Volunteer) 08/2019 - 08/2020

Supervisor : Prof. Dr. Débora Peres Menezes.

Theme: Quark stars and thermodynamic consistent models.

Published Work: Article [\[7\]](#).

Department of Chem. Eng. Research Internship 08/2016 - 08/2019

Supervisor : Prof. Dr. Ricardo Antônio Francisco Machado.

Theme : Biomass valorization and RNA sequential analysis.

Published Work: Articles [\[8, 9, 10, 11\]](#).

Articles

In preparation:

- [1] E. Hafemann, “A low-regularity Riemannian positive mass theorem for non-spin manifolds with distributional curvature”, [In preparation](#), 2025

Published:

- [2] E. Hafemann and E.-A. Kontou, “Penrose inequality for integral energy conditions”, [Classical and Quantum Gravity](#) **42**, 195016 (2025)
- [3] M. Calisti, M. Graf, E. Hafemann, M. Kunzinger, and R. Steinbauer, “Hawking’s singularity theorem for Lipschitz Lorentzian metrics”, [Commun. Math. Phys.](#) **406**, 207 (2025)

Articles in Applied Fields

Inverse Problems

- [4] A. De Cezaro, E. Hafemann, A. Leitão, and A. Osses, “A regularization method based on level-sets for the problem of crack detection from electrical measurements”, [Inverse Problems](#) **39**, 035009 (2023)
- [5] R. Filippozzi, E. Hafemann, J. C. Rabelo, F. Margotti, and A. Leitão, “A range-relaxed criteria for choosing the Lagrange multipliers in the Levenberg–Marquardt–Kaczmarz method for solving systems of non-linear ill-posed equations: Application to EIT-CEM with real data”, [Journal of Inverse and Ill-posed Problems](#) **31**, 267–292 (2023)
- [6] F. Margotti and E. Hafemann, “Range-relaxed strategy applied to the Levenberg–Marquardt method with uniformly convex penalization term in Banach spaces”, [Inverse Problems](#) **38**, 095001 (2022)

Particle Physics

- [7] B. C. Backes, E. Hafemann, I. Marzola, and D. P. Menezes, “Density-dependent quark mass model revisited: thermodynamic consistency, stability windows and stellar properties”, [Journal of Physics G: Nuclear and Particle Physics](#) **48**, 055104 (2021)

Chemical Engineering

- [8] T. Neitzel, C. S. Lima, E. Hafemann, D. A. A. Paixão, J. M. Junior, G. F. Persinoti, L. V. dos Santos, and J. L. Ienczak, “RNA-seq based transcriptomic analysis of the non-conventional yeast *Spathaspora passalidarum* during Melle-boinot cell recycle in xylose-glucose mixtures”, [Renewable Energy](#) **201**, 486–498 (2022)
- [9] E. Hafemann, R. Battisti, D. Bresolin, C. Marangoni, and R. A. F. Machado, “Enhancing chlorine-free purification routes of rice husk biomass waste to obtain cellulose nanocrystals”, [Waste and Biomass Valorization](#) **11**, 6595–6611 (2020)

[10] E. Hafemann, R. Battisti, C. Marangoni, and R. A. Machado, “Valorization of royal palm tree agroindustrial waste by isolating cellulose nanocrystals”, [Carbohydrate Polymers](#) **218**, 188–198 (2019)

[11] R. Battisti, E. Hafemann, C. A. Claumann, R. A. F. Machado, and C. Marangoni, “Synthesis and characterization of cellulose acetate from royal palm tree agroindustrial waste”, [Polymer Engineering and Science](#) **59**, 891–898 (2018)

Books

Inverse Problems:

[12] F. Margotti, E. Hafemann, and L. M. Santana, *Implementação computacional da tomografia por impedância elétrica (Computational implementation of electrical impedance tomography)*, ISBN 978-85-244-0535-8 (Editora do IMPA, 2023)

Training & Research Stay

Workshop, Masterclass and Meeting

Masterclass, *Geometrical Aspects of Mathematical Relativity*, University of Copenhagen, 16–20 June 2025.

Workshop, *Introductory Workshop – New Frontiers in Curvature*, SLMATH, Berkeley, CA, 26–30 August 2024. Travel support of \$1,000 provided by the event.

Meeting, *14th Central European Relativity Seminar*, University of Tübingen, 14–16 February 2024.

Research Stay

King’s College London with Prof. Eleni-Alexandra Kontou, 29 April – 3 May 2024 and 1–5 December 2025.

Universität Wien with Profs. Michael Kunzinger and Roland Steinbauer, 11–15 December 2023.

Talks & Presentations

Talk at Gravity Journal Club, *King’s College London*, , 1 December 2025 [2].

Poster presentation at Quantum Universe Day, *DESY*, 16 February 2025 [2].

Talk at 15th Central European Relativity Seminar, *Radboud University Nijmegen*, 22–24 January 2025 [1].

Talk at *Mathematical Relativity Seminar*, *Universität Hamburg*, 14 December 2024 and 30 April 2025 [1, 2].

Presenter at the *34th Brazilian Mathematics Colloquium (CBM)*: Advanced Course on Electrical Impedance Tomography (EIT), with collaborator Fábio Margotti. *Instituto Nacional de Matemática Pura e Aplicada (IMPA)*, 24–28 July 2023 [12].

Teaching Experience

Differential Geometry Teaching Assistant 08/2022 – 12/2022

Assisted undergraduate mathematics students with homework and review sessions.

Theory of Distributions Teaching Assistant 10/2024 – 03/2025

Responsible for creating exercise sheets and exercise classes.

Language & Skills

Languages

Brazilian-Portuguese (Native), English (Fluent), German (A1).

Computer Skills

Python, R, MATLAB, Linux, Docker, Git, Django, HTML, CSS, JavaScript, Bootstrap, FEniCS project.

Piano

I studied piano seriously for five years starting at age 14, and the creativity and discipline of music continue to inspire my research today.

References

Prof. Dr. Melanie Graf

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Prof. Dr. Eleni-Alexandra Kontou

Lecturer in Theoretical Physics, King's College London.

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Prof. Dr. Ivan Pontual Costa e Silva

Professor of Department of Mathematics, Federal University of Santa Catarina.

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